

Q1.

A small, hollow, plastic ball of mass m kg is at rest on a point O on a polished horizontal surface. The ball is attached to two identical springs. The other ends of the springs are attached to the points P and Q which are 1.8 metres apart on a straight line through O .

The ball is struck so that it moves away from O , towards P with a speed of 0.75 m s^{-1} .

As the ball moves, its displacement from O is x metres at time t seconds after the motion starts.

The force that each of the springs applies to the ball is $12.5mx$ newtons towards O .

The ball is to be modelled as a particle. The surface is assumed to be smooth and it is assumed that the forces applied to the ball by the springs are the only horizontal forces acting on the ball.

- (a) Find the minimum distance of the ball from P , in the subsequent motion.

(5)

- (b) In practice the minimum distance predicted by the model is incorrect.

Is the minimum distance predicted by the model likely to be too big or too small?

Explain your answer with reference to the model.

(2)

(Total 7 marks)

Q2.

A particle P , of mass m kg, moves in a straight horizontal line. At time t seconds, the displacement of P from a fixed point O on the line is x metres, and P is moving with velocity $\dot{x} \text{ m s}^{-1}$. Throughout the motion, two horizontal forces act on P : a force of magnitude $4mn^2|x|$ newtons directed towards O , and a resistance force of magnitude $2mk|\dot{x}|$ newtons, where n and k are positive constants.

- (a) Show that $\ddot{x} + 2k\dot{x} + 4n^2x = 0$.

(3)

- (b) In one case, $k = n$. When $t = 0$, $x = a$ and $\dot{x} = 0$.

(i) Show that $x = e^{-nt} \left(a \cos \sqrt{3}nt + \left(\frac{\sqrt{3}a}{3} \right) \sin \sqrt{3}nt \right)$.

(7)

- (ii) Show that P passes through O when $\tan \sqrt{3} nt = -\sqrt{3}$.

(3)

- (c) In a different case, $k = 2n$.

- (i) Find a general solution for x at time t seconds.

(3)

- (ii) Hence state the type of damping which occurs.

(1)

(Total 17 marks)

Q3.

A particle moves in a straight line with simple harmonic motion such that its displacement at time t seconds relative to a fixed origin on this line is x metres. The motion of the particle satisfies the differential equation

$$\frac{d^2x}{dt^2} + 16x = 0$$

- (a) Verify that

$$x = A \cos 4t + B \sin 4t$$

where A and B are constants, is a solution to this differential equation.

(4)

- (b) When $t = 0$, the particle is momentarily at rest. Show that $B = 0$.

(1)

- (c) Given that $x = h$ ($h > 0$) when $t = \frac{\pi}{12}$, find A in terms of h .

(2)

- (d) Find the maximum speed of the particle in terms of h .

(1)

- (e) The mass of the particle is m kg. Find the magnitude of the maximum force acting on the particle during the motion. Give your answer in terms of h and m .

(2)

(Total 10 marks)

Q4.

A particle, of mass 2 kg, is suspended from a fixed point O by a light spring of natural length 0.5 metres and modulus of elasticity 49 N.

- (a) Initially, the particle hangs at rest in equilibrium below O . Find the extension of the spring in this position.

(2)

- (b) A force, F newtons, is then applied to the particle in a vertically downwards direction. The displacement of the particle below its equilibrium position at time t seconds later is x metres. Given that $F = 12 \cos nt$, where n is a positive constant, show that

(5)

- (c) In the case where $n = 5$, find an expression for x at time t .

(10)

- (d) State the value of n for which resonance occurs.

(1)

(Total 18 marks)

Q5.

The ends of a light, uniform elastic string are fixed to two points, A and B , a distance $9a$ apart on a smooth, horizontal plane. The string is of natural length $6a$ and modulus of elasticity $4mn^2a$, where n is a constant.

A particle of mass m is attached to the string at P , where $AP = 6a$. The natural length of AP is $4a$ and the natural length of BP is $2a$. In this position, the particle is in equilibrium.



The particle is moved a distance $\frac{1}{2}a$ towards B and then released from rest at time $t = 0$. The displacement of the particle from its equilibrium position at time t is x .

Hence initially $x = +\frac{1}{2}a$.

The motion of the particle is resisted by a force of magnitude $2mnv$, where v is the speed of the particle at time t .

©2025 Exam Papers Practice. All rights reserved.

- (a) Show that x satisfies

$$\frac{d^2x}{dt^2} + 2n \frac{dx}{dt} + 3n^2x = 0$$

(7)

- (b) Given that $n = 1$, find x in terms of a and t .

(8)

(Total 15 marks)

Q6.

A particle P , of mass $2m$, is suspended from a fixed point O by a light elastic string of natural length a and modulus of elasticity $8mn^2a$, where n is a positive constant. The particle is released from rest when $t = 0$ at a point A , where A is vertically below O and $OA = a$.

When the particle is moving with speed v , it experiences air resistance of magnitude $4mnv$.

- (a) The displacement of P below A at time t is x . Show that x satisfies the equation

$$\frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 4n^2x = g$$

(3)

- (b) Find x in terms of n , g and t .

(12)

- (c) Hence show that the first time the particle comes to rest is when $t = \frac{\pi}{\sqrt{3}n}$.

(4)

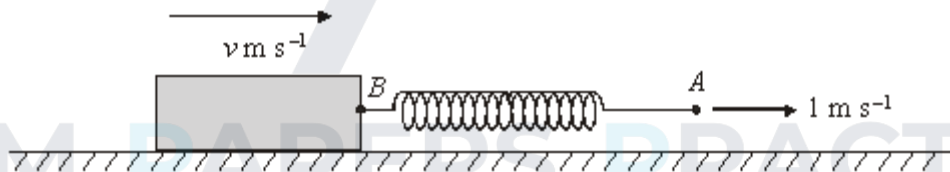
(Total 19 marks)

Q7.

A block of mass 0.5 kg moves on a smooth horizontal surface. It is attached to the end B of a light elastic spring AB , where AB is horizontal.

The natural length of the spring is 0.5 m and its stiffness is 5 N m^{-1} .

The end A of the spring is pulled horizontally with constant speed of 1 m s^{-1} , causing the block to move across the horizontal surface.



At time t seconds, the length of the spring is x metres and the block is moving with speed $v \text{ m s}^{-1}$.

The motion of the block is subject to a resistive force of $v \text{ N}$.

- (a) Write down an expression for the tension in the spring in terms of x .

(2)

- (b) Show that the equation of motion of the block is

$$\frac{dv}{dt} + 2v - 10x = -5.$$

(3)

- (c) (i) Explain why $\dot{x} = 1 - v$.

(1)

- (ii) Hence show that

$$\ddot{x} + 2\dot{x} + 10x = 7.$$

(2)

- (d) Find the general solution of this equation.

(6)

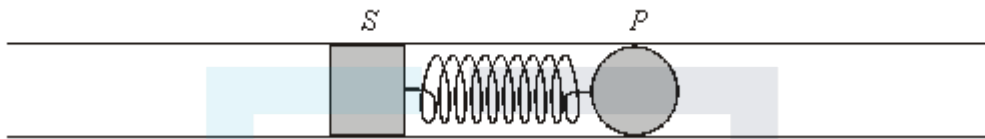
- (e) State the limiting value of x . Give a reason for your answer.

(2)

(Total 16 marks)

Q8.

A solid ball P is attached to one end of a light elastic spring. The other end of the spring is attached to a piston S which is free to move horizontally inside a smooth tube, as shown in the diagram.



The mass of P is 0.5 kg. The spring has natural length 0.5 m and stiffness 4.5 N m^{-1} . The system initially rests in equilibrium.

The piston S is forced to oscillate, causing P to move. At time t seconds, the

displacements of S and P are $\frac{1}{2} \sin 2t$ metres and x metres respectively, measured in the same direction from their initial positions.

- (a) Draw a diagram showing the forces acting on P and the displacements of P and S .

(1)

- (b) Show that x satisfies the differential equation

©2025 Exam Papers Practice. All rights reserved.

$$\ddot{x} + 9x = \frac{9}{2} \sin 2t.$$

(4)

- (c) Find x as a function of t .

(9)

- (d) Show that P is instantaneously at rest when $t = \frac{2\pi}{5}$.

(2)

(Total 16 marks)

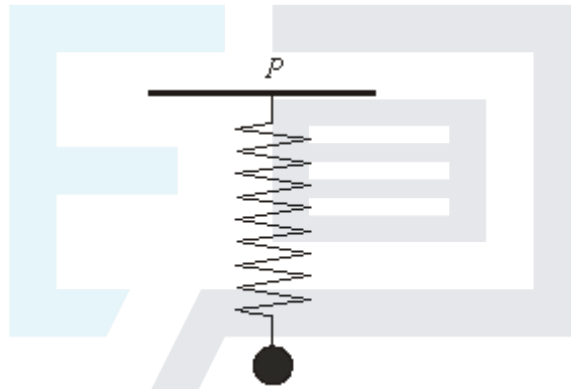
Q9.

The pointer on a set of bathroom scales oscillates before coming to rest at its final reading. The angular displacement of the pointer is θ radians at time t seconds. The oscillation of the pointer is modelled by the differential equation:

- (a) Find the general solution of this differential equation. (6)
- (b) Given that $\theta = 0.2$ and $\dot{\theta} = 0$ when $t = 0$, find the particular solution of the differential equation. (5)
- (c) State, with a reason, the limiting value of θ given by this model. (2)
- (Total 13 marks)**

Q10.

A sphere of mass 2 kg is fixed to one end of a spring. The other end of the spring is attached to the fixed point P , as shown in the diagram. The natural length of the spring is 0.5 metres.



- (a) When the system is in equilibrium, the length of the spring is 0.7 metres. Find the modulus of elasticity of the spring. (3)
- (b) The sphere is pulled down below its equilibrium position and released from rest. At time t seconds, the displacement of the sphere from its equilibrium position is x metres.
- (i) Show that
- $$\frac{d^2x}{dt^2} = -49x$$
- (5)
- (ii) Find the value of t when the sphere returns to its equilibrium position for the first time. (4)

(Total 12 marks)

Q11.

The motion of a particle of unit mass, moving along a straight line through O , is governed by the differential equation



$$\frac{d^2x}{dt^2} + 6 \frac{dx}{dt} + 10x = 325 \sin 3t$$

where x is the displacement from O at time t .

- (a) Given that the particle starts from rest at O , solve the equation.

(11)

- (b) Describe fully the nature of the motion for large values of t .

(3)

(Total 14 marks)

Q12.

An elastic string has natural length l and modulus of elasticity λ . One end of the string is fixed and a particle of mass m is attached to the other end of the string.

- (a) When the particle hangs vertically in equilibrium, the extension of the string is $\frac{l}{4}$. Show that $\lambda = 4mg$.

(2)

- (b) The particle is then set into vertical motion and x is the displacement of the particle from the equilibrium position at time t . Assume that the string remains taut during the vertical motion.

(i) Show that $\frac{d^2x}{dt^2} = \frac{4gx}{l}$.

(5)

- (ii) Hence find the period of the motion.

(2)

(Total 9 marks)

©2025 Exam Papers Practice. All rights reserved.

Q13.

A particle P , of mass m , is suspended from a fixed point O by a light elastic string of natural length $2a$ and modulus $4mn^2a$, where n is a positive constant. The particle is released from rest at a point C , where C is vertically below O and $OC = 2a$.

When the particle is moving with velocity v , it experiences air resistance of magnitude $2mnv$.

- (a) The displacement of P below C at time t is x .

Show that x satisfies the equation

$$\frac{d^2x}{dt^2} + 2n \frac{dx}{dt} + 2n^2x = g$$

(4)

- (b) Find x in terms of n , g and t .

Q14.

A spring, of natural length $4a$ and modulus λ , has one end attached to a fixed support A , and a particle P of mass m is attached to its other end. Another spring, of natural length $2a$ and modulus $4mg$, has one end attached to P and the other end attached to a fixed support B , which is situated at a distance of $10a$ vertically below A . The system is in equilibrium in a vertical line with the upper spring stretched to a length of $7a$ and the lower spring stretched to a length of $3a$ as shown in the diagram.



- (a) Show that $\lambda = 4mg$.

(4)

- (b) At time $t = 0$, the particle is lowered to a distance $\frac{a}{2}$ below its equilibrium position and released from rest. The subsequent motion of P is subject to a resistance of magnitude $\frac{1}{5}mkv$, where $k^2 = \frac{6g}{a}$ and v is the speed of the particle at time t .

- (i) Given that x is the downward displacement of P from its equilibrium position at time t , show that

$$10 \frac{d^2x}{dt^2} + 2k \frac{dx}{dt} + 5k^2x = 0$$

(6)

- (ii) Hence find x in terms of a , k and t .

(9)

(Total 19 marks)

Q15.

A particle moves along a straight line. At time t , the displacement of the particle is x . Its motion is governed by the differential equation

$$\frac{d^2x}{dt^2} + \frac{dx}{dt} = k \sin 2t$$

where k is a positive constant. The particle starts from rest at the point $x = a$.

- (a) Determine x in terms of a , k and t subject to the given initial conditions. (11)
- (b) Find the values between which x oscillates if t is so large that e^{-t} can be neglected. (4)
- (Total 15 marks)

Q16.

A particle, of mass m , is attached to one end of a spring. The spring has natural length l and modulus of elasticity λ . The other end of the spring is fixed to the point Q . Initially the particle is in equilibrium, vertically below Q .

- (a) Find, in terms of m , g , λ and l , the extension of the spring when the particle is in equilibrium. (2)
- (b) The particle is then pulled down, so that it is vertically below its equilibrium position, and released. The displacement of the particle from its equilibrium position is x at time t . Show that
- $$\frac{d^2 x}{dt^2} = -\frac{\lambda x}{ml} \quad (5)$$
- (c) Find, in terms of A , l and m , the period of the subsequent motion. (2)
- (d) The spring is now replaced with another spring of natural length l , but with modulus of elasticity $\frac{\lambda}{4}$. What happens to the period of the motion? (2)

(Total 11 marks)

©2025 Exam Papers Practice. All rights reserved.

Q17.

A particle P of mass m is suspended from a fixed point O by a light elastic string of natural length a and modulus $6mn^2a$, where n is a positive constant. The particle is released from rest at a point A , where A is vertically below O and distance OA is a .

When the particle is moving with velocity $v \text{ m s}^{-1}$ it experiences air resistance of magnitude $2mnv \text{ N}$.

- (a) The displacement of P below A at time t is x .

Show that x satisfies the equation

$$\frac{d^2 x}{dt^2} + 2n \frac{dx}{dt} + 6n^2 x = g$$

(3)

- (b) Find x in terms of n , g and t .

(10)

- (c) Hence show that the first time the particle comes to rest is when $t = \frac{\pi}{\sqrt{5n}}$.

(4)

(Total 17 marks)

Q18.

A particle moves on a straight line. At time t the displacement of the particle from a point O is x . The particle moves with simple harmonic motion, so that

$$\frac{d^2x}{dt^2} = -25x$$

The particle is initially at rest at the point P , which is a distance a from O .

- (a) Find the time that it takes the particle to move directly from P to O .

(3)

- (b) Find, in terms of a , the distance of the particle from O when its speed is half its maximum speed.

(5)

(Total 8 marks)

Q19.

The motion of a particle, which moves along a straight line, is governed by the differential equation

$$2 \frac{d^2x}{dt^2} + 6 \frac{dx}{dt} + 9x = 0$$

where x is the displacement of the particle from the origin at time t .

Initially the particle is at rest at the point where $x = a$.

- (a) Find x in terms of a and t .

(9)

- (b) Is the damping of the motion of the particle light, critical or heavy? Give a reason for your answer.

(2)

(Total 11 marks)